

1.Creating and Running Simple C Programs:

a) C-Program to calculate the sum of three numbers

```
// C Program to Find Sum of 3 Numbers
#include <stdio.h>
int main(){
    int a, b, c, sum;

    // Asking for input
    printf("Enter 3 numbers: \n");
    scanf("%d %d %d", &a, &b, &c);
    // Calculating sum
    sum = a + b + c;
    // Displaying output
    printf("Sum = %d \n", sum);
    return 0;
}
```

Output

Enter 3 numbers:

3

5

7

Sum = 15

b) C-Program to demonstrate a Simple Calculator.

```
#include <stdio.h>

int main() {

    char op;
    double first, second;
    printf("Enter an operator (+, -, *, /): ");
    scanf("%c", &op);
    printf("Enter two operands: ");
    scanf("%lf %lf", &first, &second);

    switch (op) {
        case '+':
            printf("%.1lf + %.1lf = %.1lf", first, second, first + second);
            break;
        case '-':
            printf("%.1lf - %.1lf = %.1lf", first, second, first - second);
```

```

        break;
    case '*':
        printf("%.1lf * %.1lf = %.1lf", first, second, first * second);
        break;
    case '/':
        printf("%.1lf / %.1lf = %.1lf", first, second, first / second);
        break;
    // operator doesn't match any case constant
    default:
        printf("Error! operator is not correct");
}

return 0;
}

```

Output

```

Enter an operator (+, -, *, /): *
Enter two operands: 3
4
3.0 * 4.0 = 12.0

```

c) C-Program to calculate the area and circumference of a circle using PI as a defined constant.

```

#include <stdio.h>
#include <math.h>
#define PI 3.14159
int main()
{
    float radius, area;
    printf("Enter the radius of a circle\n");
    scanf("%f", &radius);
    area = PI*radius*radius;
    printf("Area of the circle = %.2f\n", area); // printing upto two decimal places
    return 0;
}

```

Output

```

Enter the radius of a circle 10
Area of the circle =314.16

```

d) C-Program to convert temperature given in Celsius to Fahrenheit and Fahrenheit to Celsius

```
#include <stdio.h>
int main()
{
    float celsius, fahrenheit;
    printf("Enter temperature in Celsius: ");
    scanf("%f", &celsius);
    //celsius to fahrenheit conversion formula
    fahrenheit = (celsius * 9 / 5) + 32;
    printf("%.2f Celsius = %.2f Fahrenheit", celsius, fahrenheit);
    return 0;
}
```

Output:

Enter temperature in Celsius: 10
10.00 Celsius = 50.00 Fahrenheit

e) C-Program to compute the roots of a quadratic equation by accepting the coefficients.

```
#include <stdio.h>
#include <math.h>

void main()
{
    int a,b,c,d;
    float x1,x2,real,img;

    printf("Input the value of a,b & c : ");
    scanf("%d%d%d",&a,&b,&c);
    if(a==0 &&b==0)
    {
        printf(" invalid inputs\n");
    }
    else if(a==0)
    {
        printf(" Leniar equation\n");
        x1=-c/b;
        printf(" linear equation is: %f",x1);
    }
}
```

```

    }
    else
    {
        d=b*b-4*a*c;
        if(d==0)
        {
            printf("Both roots are equal.\n");
            x1=-b/(2.0*a);
            x2=x1;
            printf("First Root Root1= %f\n",x1);
            printf("Second Root Root2= %f\n",x2);
        }
        else if(d>0)
        {
            printf("Both roots are real and diff-2\n");
            x1=(-b+sqrt(d))/(2*a);
            x2=(-b-sqrt(d))/(2*a);
            printf("First Root Root1= %f\n",x1);
            printf("Second Root root2= %f\n",x2);
        }
        else
        {
            printf("Root are imeainary\n");
            real=-b/(2*a);
            img= sqrt(fabs(d))/(2*a);
            printf("root 1:%f+%fi \n root 2:%f-%fi",real,img,real,img);
        }
    }
}

```

Output

Input the value of a,b & c : 0

0

2

invalid inputs

Input the value of a,b & c : 0

2

6

Linear equation

linear equation is: -3.000000

Input the value of a,b & c : 1

4

1

Both roots are real and distinct

First Root Root1= -0.267949

Second Root root2= -3.732051

Input the value of a,b & c : 1

2

1

Both roots are equal.

First Root Root1= -1.000000

Second Root Root2= -1.000000

Input the value of a,b & c : 2

1

1

Root are imaginary

root 1:0.000000+0.661438i

root 2:0.000000-0.661438i

2. Creating and Running C Programs on Expressions :

a) C-Program to calculate quotient and reminder of two numbers

```
#include <stdio.h>
int main() {
    int dividend, divisor, quotient, remainder;
    printf("Enter dividend: ");
    scanf("%d", &dividend);
    printf("Enter divisor: ");
    scanf("%d", &divisor);

    // Computes quotient
    quotient = dividend / divisor;

    // Computes remainder
    remainder = dividend % divisor;

    printf("Quotient = %d\n", quotient);
    printf("Remainder = %d", remainder);
    return 0;
}
```

Output

```
Enter dividend: 25
Enter divisor: 4
Quotient = 6
Remainder = 1
```

b) C-Program to evaluate two complex expressions.

```
#include<stdio.h>
int main()
{
    //local declarations
    int a=3;
    int b=4;
    int c=5;
    int x;
    int y;

    //statements
    printf("initial values of variables:\n");
    printf("a=%d\tb=%d\tc=%d\n\n",a,b,c);
    x=a*4+b/2-c*b;
```

```

printf("value of the expression a*4+b/2-c*b :%d\n",x);
y=-a*(3+b)/2-c++*b;
printf("value of the expression --a*(3+b)/2-c++*b :%d\n",y);
printf("values of variables are now:\n");
printf("a=%d\tb=%d\tc=%d\n\n",a,b,c);
return 0;
}

```

Output

initial values of variables:

a=3 b=4 c=5

value of the expression $a*4+b/2-c*b$:-6

value of the expression $--a*(3+b)/2-c++*b$:-13

values of variables are now:

a=2 b=4 c=6

c) C-Program to demonstrate automatic and type casting of numeric

```

#include<stdio.h>
#include<stdbool.h>
int main()
{
    //local declarations
    bool b=true;
    char c='A';
    float d=245.3;
    int i=3650;
    short s=78;

    //statements
    printf("bool+char is char:%c\n",b+c);
    printf("int * short is int: %d\n",i*s);
    printf("float * char is float: %f\n",d*c);

    c=c+b;
    d=d+c;
    b=false;
    b=-d;

    printf("\n After execution...\n");
    printf("char +true: %c\n",c);
}

```

```

printf("float +char :%f\n",d);
printf("bool =-float:%f\n",b);

return 0;
}

```

Output

```

bool+char is char:B
int * short is int: 284700
float * char is float: 15944.500000

```

After execution...

```

char +true: B
float +char :311.299988
bool =-float:311.299988

```

d) C-Program to calculate the total sales given the unit price, quantity, discount and tax rate

```

#include<stdio.h>
#define TAX_RATE 8.50
int main()
{
    //local declarations
    int quantity;
    float discontrate;
    float discountamt;
    float unitprice;
    float subtotal;
    float subtaxable;
    float taxam;
    float total;

    //statements
    printf("enter the number of items sold:\n");
    scanf("%d",&quantity);

    printf("enter the unit price:\n");
    scanf("%f",&unitprice);

```



```

printf("enter the discount rate(percent):\n");
scanf("%f",&discontrate);

subtotal=quantity*unitprice;
discountamt=subtotal*discontrate/100;
subtaxable=subtotal-discountamt;
taxam=subtaxable*TAX_RATE/100;
total=subtaxable+taxam;

printf("\n quantity sold:    %6d\n",quantity);
printf("unit price of items : %9.2f\n",unitprice);
printf("          -----\n");

printf("subtotal :          %9.2f\n",subtotal);
printf("discount :          %-9.2f\n",discountamt);
printf("discount total:      %9.2f\n",subtaxable);
printf("sales tax:           %+9.2f\n",taxam);
printf("Totsl sales:         %9.2f\n",total);

return 0;
}

```

Output

```

enter the number of items sold:
5
enter the unit price:
10
enter the discount rate(percent):
10
quantity sold:          5
unit price of items :   10.00
          -----
subtotal :              50.00
discount :              5.00
discount total:        45.00
sales tax:              +3.83
Totsl sales:           48.83

```

e) C-Program to calculate a student's average score for a course with 4 quizzes, 2 midterms and a final. The quizzes are weighted 30%, the midterms 40% and the final 30%.

```
#include<stdio.h>
#define QUIZ_WEIGHT 30
#define MIDTERM_WEIGHT 40
#define FINAL_WEIGHT 30
#define QUIZ_MAX 400.00
#define MIDTERM_MAX 200.0
#define FINAL_MAX 100.00

int main()
{
    int quiz1,quiz2,quiz3,quiz4,totalquiz;
    int midterm1,midterm2,totalmidterm,final;
    float quizpercent,midtermpercent,finalpercent,totalpercent;

    printf("Enter the score of first quiz:");
    scanf("%d",&quiz1);
    printf("Enter the score of second quiz:");
    scanf("%d",&quiz2);
    printf("Enter the score of third quiz:");
    scanf("%d",&quiz3);
    printf("Enter the score of fourth quiz:");
    scanf("%d",&quiz4);

    printf("enter the score of first midterm:");
    scanf("%d",&midterm1);
    printf("enter the score of seconf midterm:");
    scanf("%d",&midterm2);
    printf("enter the score of Final:");
    scanf("%d",&final);

    totalquiz=quiz1+quiz2+quiz3+quiz4;
    totalmidterm=midterm1+midterm2;

    quizpercent=(float)totalquiz*QUIZ_WEIGHT/QUIZ_MAX;
    midtermpercent=(float)totalmidterm*MIDTERM_WEIGHT/MIDTERM_MAX;
    finalpercent=(float)final*FINAL_WEIGHT/FINAL_MAX;
```

```

        totalpercent=quizpercent+midtermpercent+finalpercent;

        printf(" quiz 1:%d\n",quiz1);
        printf(" quiz 2:%d\n",quiz2);
        printf(" quiz 3:%d\n",quiz3);
        printf(" quiz 4:%d\n",quiz4);
        printf(" quiz total:%d\n",totalquiz);
        printf("midterm 1:%4d\n",midterm1);
        printf("midterm 2:%4d\n",midterm2);
        printf("midterm total:%4d\n",totalmidterm);
        printf("Final :%4d\n",final);
        printf("quiz  %6.1lf\n",quizpercent);
        printf("midterm  %6.1lf\n",midtermpercent);
        printf("final  %6.1lf\n",finalpercent);
        printf("total  %6.1lf\n",totalpercent);
        return 0;
    }

```

Output

```

Enter the score of first quiz:98
Enter the score of second quiz:89
Enter the score of third quiz:78
Enter the score of fourth quiz:79
enter the score of first midterm:90
enter the score of seconf midterm:100
enter the score of Final:92
quiz 1:98
quiz 2:89
quiz 3:78
quiz 4:79
quiz total:344
midterm 1: 90
midterm 2: 100
midterm total: 190
Final : 92
quiz  25.8
midterm  38.0
final  27.6
total  91.4

```

3 Creating and Running C Programs on Making Decision:

a) C-Program to determine the use of the character classification functions found in c-type library. (Page:267)

```
#include<stdio.h>
#include<ctype.h>

int main()
{
    char charIn;

    printf("enter the character to be examined\n");
    scanf("%c", &charIn);

    if(islower(charIn))
        printf("You have entered a lower case letter\n");
    else if(isupper(charIn))
        printf("You have entered a UPPER case letter\n");
    else if(isdigit(charIn))
        printf("You have entered a digit\n");
    else if(ispunct(charIn))
        printf("You have entered a Punctuation character\n");
    else if(isspace(charIn))
        printf("You have entered a Whitespace character\n");
    else
        printf("You have entered a control character\n");
    return 0;
}
```

Output:

enter the character to be examined
a
You have entered a lower case letter

enter the character to be examined
,
You have entered a Punctuation character

b) C-Program to read a test score, calculate the grade for the score and print the grade. (Page: 259)

```
#include<stdio.h>

int main()
{
```

```

int marks,temp;

/*C Program to Find Grade of a Student Using Switch Case*/

printf("\n-----");
printf("\nEnter The Marks Between 0 To 100:");

printf("\nEnter The Mark: ");
scanf("%d", &marks);

if(marks>100)
{
    /* Marks greater than 100 */
    printf("\nDon't Be Smart Enter your Marks Between Limit\n");
}
else
{
    temp=marks/10;
    switch(temp)
    {
        case 10 :
        case 9 :
            /* Marks between 90-100 */
            printf("\n Your Grade is: A");
            break;
        case 8 :
            /* Marks between 80-89 */
            printf("\n Your Grade is: B" );
            break;
        case 7 :
            /* Marks between 70-79 */
            printf("\n Your Grade is: C" );
            break;
        case 6 :
            /* Marks between 60-69 */
            printf("\n Your Grade is: D" );
            break;
        default :
            /* Marks less than 40 */
            printf("\n You Grade is: F or Fail\n");
    }
}

return 0;
}

```

Output:

Enter The Marks Between 0 To 100:

Enter The Mark: 55

You Grade is: F or Fail

Enter The Marks Between 0 To 100:

Enter The Mark: 90

Your Grade is: A

c) C-Program to uses a menu to allow the user to add, multiply, subtract and divide two numbers using switch case. (Page: 277)

```
#include <stdio.h>
```

```
#include<stdlib.h>
```

```
/**
```

```
 * Function declarations for calculator
```

```
 */
```

```
int getoption(void);
```

```
float calc(int option,int num1,int num2);
```

```
float add(int num1, int num2);
```

```
float sub(int num1, int num2);
```

```
float mult(int num1, int num2);
```

```
float divi(int num1, int num2);
```

```
int main()
```

```
{
```

```
    int option;
```

```
    int num1, num2;
```

```
    float result;
```

```
    option=getoption();
```

```
    printf("please enter two integer numbers:");
```

```
    scanf("%d%d",&num1,&num2);
```

```
    result=calc(option,num1,num2);
```

```
    printf("In main result is:%4.2f\n",result);
```

```
    return 0;
```

```
}
```

```
int getoption(void)
```

```

{
    int option;
    /* Print welcome message */
    printf("WELCOME TO SIMPLE CALCULATOR\n");
    printf("-----\n");
    printf("Enter [number 1] :add\n");
    printf("Enter [number 2] :Subtract\n");
    printf("Enter [number 3] :multiply\n");
    printf("Enter [number 4] :divide\n");

    /* Input two number and operator from user */
    scanf("%d", &option);
    return option;
}

```

```

float calc(int option,int num1,int num2)
{
    float result;

    switch(option)
    {
        case 1:
            result = add(num1, num2);
            break;

        case 2:
            result = sub(num1, num2);
            break;

        case 3:
            result = mult(num1, num2);
            break;

        case 4:
            if(num2==0.0)
            {
                printf("error");
                exit(0);
            }
            else
            {
                result = divi(num1, num2);
                break;
            }
        default:
            printf("Invalid operator");
    }
}

```

```

    }

    /* Print the result */
    printf("Result = %4.2f", result);

    return result;
}

/**
 * Function to add two numbers
 */
float add(int num1, int num2)
{
    return num1 + num2;
}

/**
 * Function to subtract two numbers
 */
float sub(int num1, int num2)
{
    return num1 - num2;
}

/**
 * Function to multiply two numbers
 */
float mult(int num1, int num2)
{
    return num1 * num2;
}

/**
 * Function to divide two numbers
 */
float divi(int num1, int num2)
{
    return num1 / num2;
}
Output

```

WELCOME TO SIMPLE CALCULATOR

Enter [number 1] :add
Enter [number 2] :Subtract

Enter [number 3] :multiply
Enter [number 4] :divide
1
please enter two integer numbers:2
3
Result = 5.00
In main result is:5.00

d) C-Program to read the name of the user, number of units consumed and print out the charges. An electricity board charges the following rates for the use of electricity:

For the first 200 units 80 paise per unit

For the next 100 units 90 paise per unit

Beyond 300 units Rs 1 per unit.

All users are charged a minimum of Rs. 100 as meter charge.If the total amount is more than Rs 400, then an additional surcharge of 15% of total amount is charged.

```
#include<stdio.h>
#include<string.h>

void main()
{
    int cust_no, unit_con;
    float charge,surcharge=0, amt, total_amt;
    char nm[25];

    printf("Enter the customer IDNO :\\t");
    scanf("%d",&cust_no);
    printf("Enter the customer Name :\\t");
    scanf("%s",nm);
    printf("Enter the unit consumed by customer :\\t");
    scanf("%d",&unit_con);

    if (unit_con <200 )
        charge = 0.80;
    else if (unit_con >=200 && unit_con <300)
        charge = 0.90;
    else
        charge = 1.00;

    amt = unit_con*charge;
    if (amt >400)
        surcharge = amt*15/100.0;
    total_amt = amt+surcharge;
```

```

printf("\t\t\t\nElectricity Bill\n\n");
printf("Customer IDNO          : \t%d", cust_no);
printf("\nCustomer Name          : \t%s", nm);
printf("\nunit Consumed           : \t%d", unit_con);
printf("\nAmount Charges @Rs. %4.2f per unit : \t%0.2f", charge, amt);
printf("\nSurcharge Amount         : \t%0.2f", surcharge);
printf("\nMinimum meter charge Rs      : \t%d", 100);
printf("\nNet Amount Paid By the Customer : \t%0.2f", total_amt+100);

}

```

Output

```

Enter the customer IDNO :    123
Enter the customer Name :    pallavi
Enter the unit consumed by customer : 150

```

Electricity Bill

```

Customer IDNO          : 123
Customer Name          : pallavi
unit Consumed          : 150
Amount Charges @Rs. 0.80 per unit : 120.00
Surcharge Amount       : 0.00
Minimum meter charge Rs : 100
Net Amount Paid By the Customer : 220.00

```

4. Creating and Running C Programs on Repetition or Loops:

a) C-Program to print a number series from 1 to a user-specified limit in the form of a right triangle

```
#include<stdio.h>
intmain()
{
    inti, j, rows, count = 1;
    printf("Enter the number of rows\n");
    scanf("%d", &rows);
    for (i = 0; i< rows; i++) {
        for (j = 0; j <= i; j++) {
            printf("%d ", count);
            count++;
        }
        printf("\n");
    }
    return(0);
}
```

Output:

Enter the number of rows

5

1

2 3

4 5 6

7 8 9 10

11 12 13 14 15

b) C-Program to print the number and sum of digits in an integer.

```
#include<stdio.h>
intmain()
{
    intn,sum=0,m;
    printf("Enter a number:");
    scanf("%d",&n);
    while(n>0)
    {
        m=n%10;
        sum=sum+m;
        n=n/10;
    }
}
```

```

        printf("Sum is=%d",sum);
        return 0;
    }

```

Output:

Enter a number:1234

Sum is=10

c) C-Program to calculate the factorial of a number using for loop/Recursion

```
#include<stdio.h>
```

```
intmain()
```

```

{
    inti,fact=1,number;
    printf("Enter a number: ");
    scanf("%d",&number);
    for(i=1;i<=number;i++){
        fact=fact*i;
    }
    printf("Factorial of %d is: %d",number,fact);
    return 0;
}

```

```

}
*****

```

//recursion

```
#include<stdio.h>
```

```
long factorial(int n)
```

```

{
    if (n == 0)
        return 1;
    else
        return(n * factorial(n-1));
}

```

```
void main()
```

```

{
    int number;
    long fact;
    printf("Enter a number: ");
    scanf("%d", &number);
    fact = factorial(number);
    printf("Factorial of %d is %ld\n", number, fact);
    return 0;
}

```

Output:

Enter a number: 5

Factorial of 5 is 120

d) C-Program to calculate nth Fibonacci number.

```
#include<stdio.h>
intmain()
{
int n1=0,n2=1,n3,i,number;
printf("Enter the number of elements:");
scanf("%d",&number);
printf("\n%d %d",n1,n2);//printing 0 and 1
for(i=2;i<number;++i)//loop starts from 2 because 0 and 1 are already printed
{
n3=n1+n2;
printf(" %d",n3);
n1=n2;
n2=n3;
}
return 0;
}
*****
```

//recursion

```
#include<stdio.h>
void fibonacciSeries(int n)
{
int a=0, b=1, c;
for(inti=0; i<n; i++)
{
printf("%d\t", a);
c = a+b;
a = b;
b = c;
}
}

intmain()
{
int term;
printf("Enter the term: ");
```

```

        scanf("%d", &term);
        printf("The fibonacci series is: \n");
        fibonacciSeries(term);
        return 0;
    }

```

Output:

Enter the number of elements:20

0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4181

e) C-Program to convert binary to a decimal number

```

#include <stdio.h>
void main()
{
    // declaration of variables
    int num, binary_num, decimal_num = 0, base = 1, rem;
    printf (" Enter a binary number with the combination of 0s and 1s \n");
    scanf (" %d", &num); // accept the binary number (0s and 1s)
    binary_num = num; // assign the binary number to the binary_num variable
    while ( num> 0)
    {
        rem = num % 10;
        decimal_num = decimal_num + rem * base;
        num = num / 10; // divide the number with quotient
        base = base * 2;
    }
    printf( " The binary number is %d \t", binary_num);
    printf (" \n The decimal number is %d \t", decimal_num);
}

```

Output:

Enter a binary number with the combination of 0s and 1s

0101

The binary number is 101

The decimal number is 5

5. Creating and Running C Programs on One Dimensional Arrays:

a) C-Program to print square of index and print it.

```
#include<stdio.h>
#define ARY_SIZE 5
intmain()
{
    intsqrr[ARY_SIZE];
    for(inti=0 ; i< ARY_SIZE ; i++)
        sqrr[i]=i*i;
    printf("Element\tSquare\n");
    for(inti=0 ; i< ARY_SIZE ; i++)
        printf("%5d\t%4d\n" , i, sqrr[i]);
    return 0;
}
```

Output:

Element Square

0	0
1	1
2	4
3	9
4	16

b) C-Program to calculate average of the numbers in an array.

```
#include <stdio.h>
```

```
intmain()
{
    intarray[10] = {1, 2, 3, 4, 5, 6, 7, 8, 9, 0};
    int sum, loop;
    float avg;
    sum = avg = 0;

    for(loop = 0; loop < 10; loop++)
    {
        sum = sum + array[loop];
    }
    avg = (float)sum / loop;
    printf("Average of array values is %.2f", avg);
}
```

```
        return 0;
    }
```

Output:

Average of array values is 4.50

c) C-Program to sort the list using bubble sort.

```
#include <stdio.h>
```

```
intmain()
```

```
{
```

```
    intarray[100], n, i, j, swap;
```

```
    printf("Enter number of elements\n");
```

```
    scanf("%d", &n);
```

```
    printf("Enter %d integers\n", n);
```

```
    for (i = 0; i < n; i++)
```

```
        scanf("%d", &array[i]);
```

```
    for (i = 0 ;i < n - 1; i++)
```

```
    {
```

```
        for (j = 0 ;j < n - i - 1; j++)
```

```
        {
```

```
            if (array[j] > array[j+1])
```

```
            {
```

```
                swap    = array[j];
```

```
                array[j] = array[j+1];
```

```
                array[j+1] = swap;
```

```
            }
```

```
        }
```

```
    }
```

```
    printf("Sorted list in ascending order:\n");
```

```
    for (i = 0; i < n; i++)
```

```
        printf("%d\n", array[i]);
```

```
    return 0;
```

```
}
```

Output:

Enter number of elements

5

Enter 5 integers

6

5

3

8
1
1
3
5
6
8

Sorted list in ascending order:

d) C-Program to search an ordered list using binary search.

```
#include <stdio.h>
int binarySearch(int array[], int x, int low, int high)
{
    // Repeat until the pointers low and high meet each other
    while (low <= high)
    {
        int mid = low + (high - low) / 2;
        if (array[mid] == x)
            return mid;
        if (array[mid] < x)
            low = mid + 1;
        else
            high = mid - 1;
    }
    return -1;
}
int main(void)
{
    int array[] = {3, 4, 5, 6, 7, 8, 9};
    int n = sizeof(array) / sizeof(array[0]);
    int x = 4;
    int result = binarySearch(array, x, 0, n - 1);
    if (result == -1)
        printf("Not found");
    else
        printf("Element is found at index %d", result);
    return 0;
}
```

Output: Element is found at index 1

6. Creating and Running C Programs on Two Dimensional Arrays:

a) C-Program to perform addition of two matrices.

```
#include <stdio.h>
```

```
int main()
{
    int m, n, c, d, first[10][10], second[10][10], sum[10][10];
    printf("Enter the number of rows and columns of matrix\n");
    scanf("%d%d", &m, &n);
    printf("Enter the elements of first matrix\n");

    for (c = 0; c < m; c++)
        for (d = 0; d < n; d++)
            scanf("%d", &first[c][d]);
    printf("Enter the elements of second matrix\n");
    for (c = 0; c < m; c++)
        for (d = 0; d < n; d++)
            scanf("%d", &second[c][d]);
    printf("Sum of entered matrices:-\n");
    for (c = 0; c < m; c++) {
        for (d = 0; d < n; d++) {
            sum[c][d] = first[c][d] + second[c][d];
            printf("%d\t", sum[c][d]);
        }
        printf("\n");
    }
    return 0;
}
```

Output:

Enter the number of rows and columns of matrix

2

2

Enter the elements of first matrix

1

1

1

1

Enter the elements of second matrix

1

1

1

1

Sum of entered matrices: -

2 2

2 2

b) C-Program to perform multiplication of two matrices.

```
#include <stdio.h>
```

```
intmain()
```

```
{
```

```
    int m, n, c, d, first[10][10], second[10][10], sum[10][10];
```

```
    printf("Enter the number of rows and columns of matrix\n");
```

```
    scanf("%d%d", &m, &n);
```

```
    printf("Enter the elements of first matrix\n");
```

```
    for (c = 0; c < m; c++)
```

```
        for (d = 0; d < n; d++)
```

```
            scanf("%d", &first[c][d]);
```

```
    printf("Enter the elements of second matrix\n");
```

```
    for (c = 0; c < m; c++)
```

```
        for (d = 0 ; d < n; d++)
```

```
            scanf("%d", &second[c][d]);
```

```
    printf("Sum of entered matrices:-\n");
```

```
    for (c = 0; c < m; c++) {
```

```
        for (d = 0 ; d < n; d++) {
```

```
            sum[c][d] = first[c][d] + second[c][d];
```

```
            printf("%d\t", sum[c][d]);
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```

Output:

Enter the number of rows and columns of matrix

3

3

Enter the elements of first matrix

1 2 3

4 5 6

7 8 9

Enter the elements of second matrix

1 1 1

1 1 1

1 1 1

Sum of entered matrices:-

2 3 4

5 6 7

8 9 10

c) C-Program to find transpose of the given matrices.

```
#include <stdio.h>
```

```
intmain()
```

```
{
```

```
    int m, n, c, d, matrix[10][10], transpose[10][10];
```

```
    printf("Enter the number of rows and columns of a matrix\n");
```

```
    scanf("%d%d", &m, &n);
```

```
    printf("Enter elements of the matrix\n");
```

```
    for (c = 0; c < m; c++)
```

```
        for (d = 0; d < n; d++)
```

```
            scanf("%d", &matrix[c][d]);
```

```
    for (c = 0; c < m; c++)
```

```
        for (d = 0; d < n; d++)
```

```
            transpose[d][c] = matrix[c][d];
```

```
    printf("Transpose of the matrix:\n");
```

```
    for (c = 0; c < n; c++) {
```

```
        for (d = 0; d < m; d++)
```

```
            printf("%d\t", transpose[c][d]);
```

```
            printf("\n");
```

```
        }
```

```
    return 0;
```

```
}
```

Output:

Enter the number of rows and columns of a matrix

2

2

Enter elements of the matrix

2 5

6 8

Transpose of the matrix:

2 6

5 8

d) C-Program to find row sum and column sum and sum of all elements in a matrix.

```
#include <stdio.h>
```

```
intmain()
```

```
{
```

```
    intm,n , sum=0;    //Row Column Declaration
```

```
    printf("Enter the number of rows and column\n");
```

```
    scanf("%d %d",&m,&n);    //Row Column Initialization
```

```
    intarr[m][n];    //Matrix Declaration
```

```
    printf("Enter the elements of the matrix\n");
```

```
    for(inti=0;i<m;i++)    //Matrix Initialization
```

```
    {
```

```
        for(int j=0;j<n;j++)
```

```
        {
```

```
            scanf("%d",&arr[i][j]);
```

```
            sum= sum+arr[i][j]
```

```
        }
```

```
    }
```

```
    printf("\nElements in the matrix are \n");
```

```
    for(inti=0;i<m;i++)    //Print Matrix
```

```
    {
```

```
        for(int j=0;j<n;j++)
```

```
        {
```

```
            printf("%d ",arr[i][j]);
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    printf("\nRow Sum....\n");
```

```
    for(inti=0;i<m;i++)
```

```
    {
```

```
        intrsum=0;
```

```
        for(int j=0;j<n;j++)
```

```
        {
```

```
            rsum=rsum+arr[i][j];
```

```

    }
    printf("\nSum of all the elements in row %d is %d\n",i,rsum);
}
printf("\nColumn Sum....\n");
for(int i=0;i<m;i++)
{
    int csum=0;
    for(int j=0;j<n;j++)
    {
        csum=csum+arr[j][i];
    }
    printf("\nSum of all the elements in column %d is %d\n",i,csum);
}
printf("Sum of all elements=%d", sum);
return 0;
}

```

Output:

Enter the number of rows and column

3

3

Enter the elements of the matrix

1 2 3

4 5 6

7 8 9

Elements in the matrix are

7 8 9

7 8 9

7 8 9

Row Sum....

Sum of all the elements in row 0 is 24

Sum of all the elements in row 1 is 24

Sum of all the elements in row 2 is 24

Column Sum....

Sum of all the elements in column 0 is 21
Sum of all the elements in column 1 is 24
Sum of all the elements in column 2 is 27
Sum of all elements=45

e)C-Program initialize/fill all the diagonal elements of a matrix with zero and print.

```
#include <stdio.h>
```

```
#define MAX 100
```

```
intmain()
```

```
{
```

```
    int n = 3, m = 3;
```

```
    int mat[][MAX] = { { 2, 1, 7 },  
    { 3, 7, 2 },  
    { 5, 4, 9 } };
```

```
    for (inti = 0; i < n; i++) {  
        for (int j = 0; j < m; j++) {
```

```
        // right and left diagonal condition  
        if (i == j || (i + j + 1) == n)  
            mat[i][j] = 0;  
    }
```

```
    }  
    for (inti = 0; i < n; i++) {  
        for (int j = 0; j < m; j++)  
            printf("%d ", mat[i][j]);  
        printf("\n");
```

```
    }  
    return 0;
```

```
}
```

Output:

0 1 0

3 0 2

0 4 0